

TORSION

FUNDAMENTAL MODE OF
LOADING

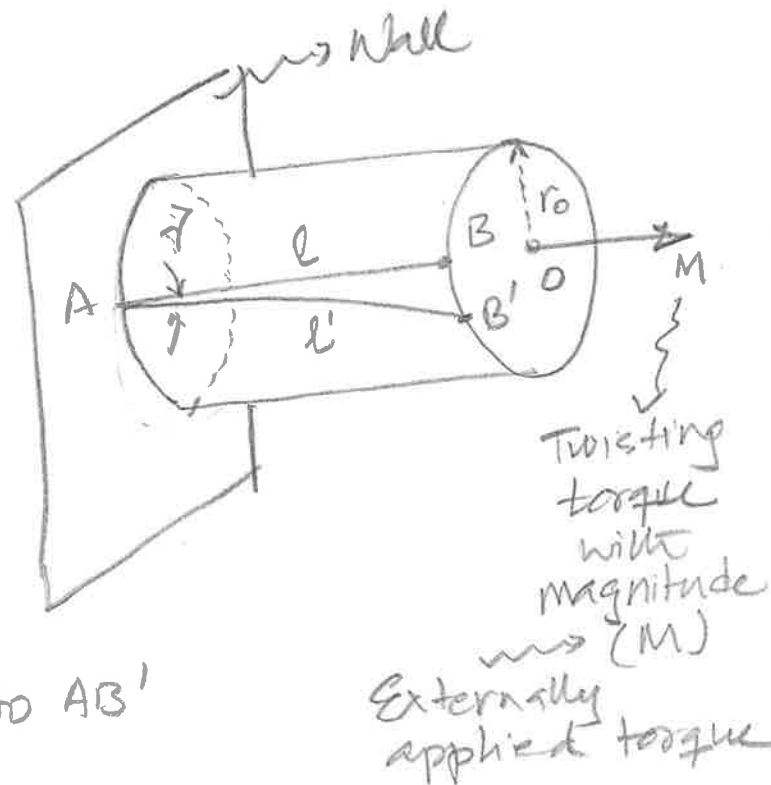
RESULTING FROM TWISTING
ACTION OF APPLIED FORCES

① CONSIDER A SOLID CIRCULAR SHAFT

① Length (l)

① Radius (r_0)

① AB is a straight
line on the outer
surface of the
shaft $\parallel$ to the
centerline



① AB is "twisted" to AB'

① Deformation @ A is ZERO.

① Deformation increases from fixed end to free end.

① Angle γ is a measure of deformation.

$$\tan \gamma \propto \frac{\text{arc length } BB'}{l}$$

(2)

For small deformations

$$\tan \gamma \approx \gamma$$

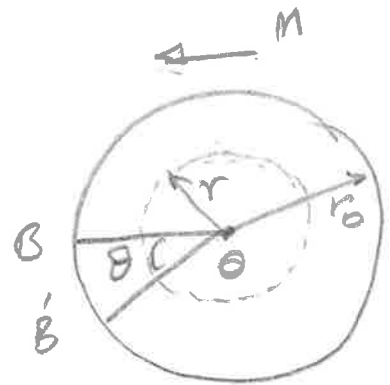
$$\Rightarrow \gamma = \frac{\text{arc length } BB'}{l}$$

→ (1)

{ Along axial direction }

(#) Amount of deformation within shaft

→ varies w.r.t radial distance (r) from the center-line of the shaft.



→ deformation ZERO at the center ($r=0$)
 → deformation increases toward rim
 → deformation maximum on outer surface ($r=r_0$)

For small deformations

$$\theta = \frac{\text{arc length } BB'}{r_0}$$

$\theta \rightarrow$ Angle of twist

→ (2)

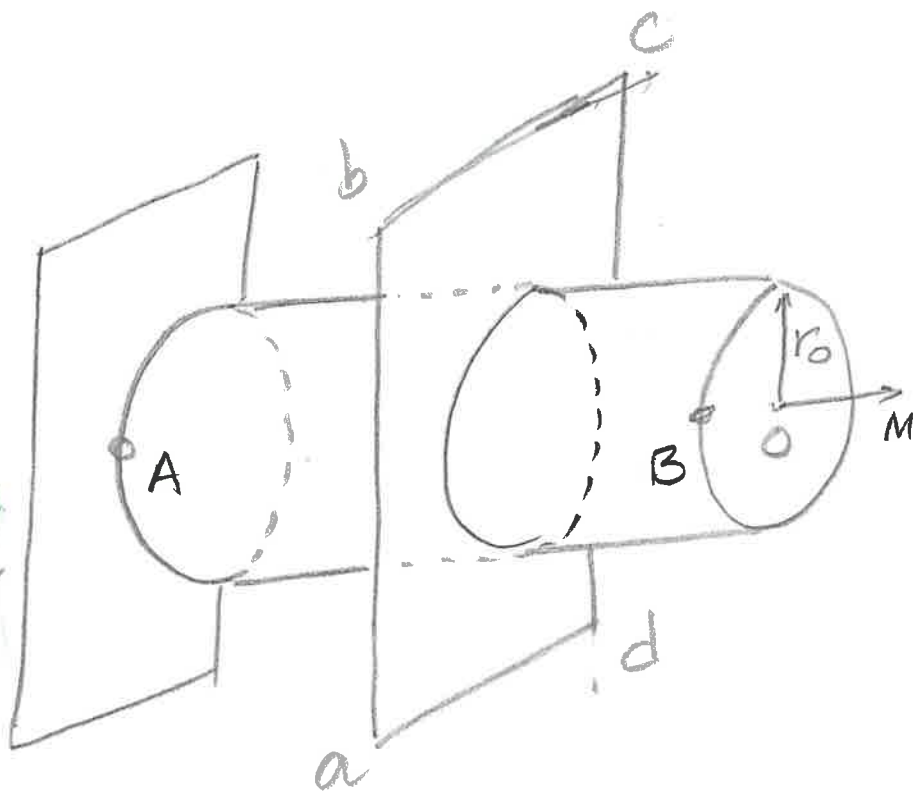
Combine equations (1) & (2)

$$\theta = \frac{r}{r_0} \gamma$$

→ (3)

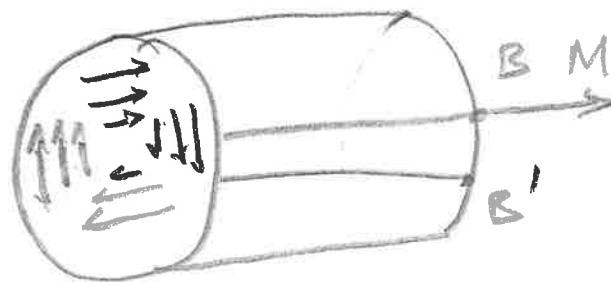
angle of twist

- # Consider a plane perpendicular to the centerline of the shaft



- # Since the whole shaft is in static equilibrium, individual parts have to be in static equilibrium

⇒ internal shear forces are distributed along cross-sectional area



⇒ { Shear force intensity } = Shear stress

(shear stress) $\propto$ (Magnitude of torque applied) (4)

$\left. \begin{array}{l} \text{Torsion} \\ \text{formula} \end{array} \right\} \tau = \frac{Mr}{J}$
Torque $\rightarrow$ (4)
shear stress $\rightarrow$
polar moment of inertia $\rightarrow$

τ_{ro} for solid shaft $I = \frac{\pi r_o^4}{2}$
 τ_{ri} for Hollow shaft $I = \frac{\pi (r_o^4 - r_i^4)}{2}$

For Hookean LEHI:

$\tau = G \gamma$ $\rightarrow$ (5)
shear stress $\rightarrow$
shear modulus $\rightarrow$

From (4) & (5)

$\gamma = \frac{\tau}{G} = \frac{Mr}{JG}$ $\rightarrow$ (6)

At $r = r_o$

$\gamma = \frac{Mr_o}{JG}$ $\rightarrow$ (7)

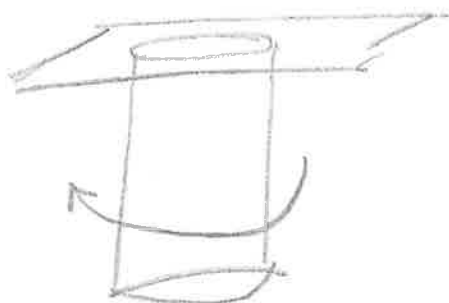
Substituting in (3)
 $\theta = \frac{L\gamma}{r_o}$

(Practical) Angle of twist

$\theta = \frac{Ml}{JG}$ $\rightarrow$ (8)

(5)

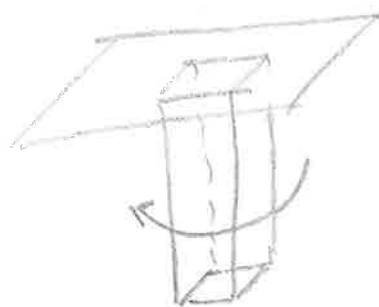
Derivations



$u_\theta \rightarrow$ exists

$u_z \rightarrow$ uniform for any cross-section

} no warping



NOT CONSIDERED

$u_\theta \rightarrow$ exists

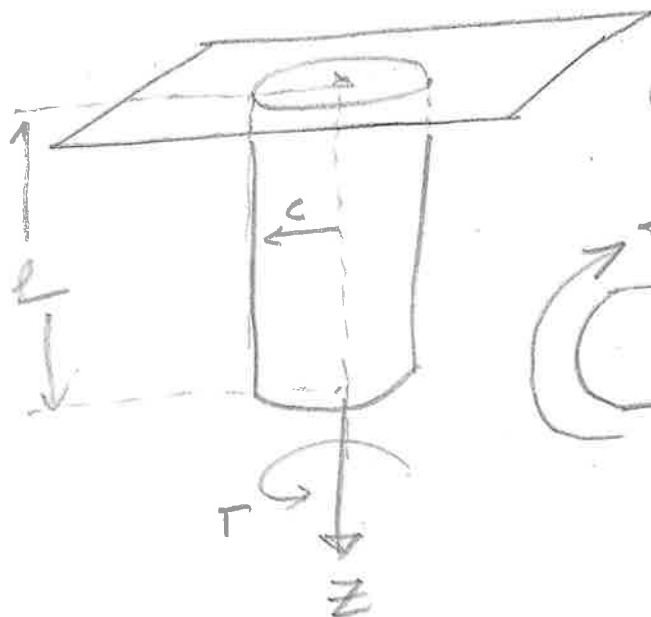
$u_z \rightarrow$ non-uniform

} warping

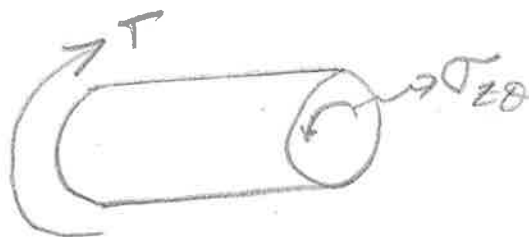


Consider a simplest case:

- ① Small twist
- ② Circular cross-sections (hollow or solid)
- ③ Hookean LEHI



(b) Consider a circular member fixed at one end



$$dA = r d\theta dr$$

(*) Free end is subject to a torque $T (\equiv M_z)$

(*) The whole system is in static equilibrium
 $\Rightarrow \therefore$ equal & opposite Torque (T) on the fixed end

(*) Equilibrium of parts

$\Rightarrow \tau_{z\theta}$ must balance T on the cut surface

$$\Rightarrow \sum M_z = 0$$

$$\Rightarrow \left[T - \int r \tau_{z\theta} dA = 0 \right] \rightarrow (*)$$

$\swarrow$ arm $\swarrow$ differential area $\swarrow$ $d\theta$

Notes: $\tau_{z\theta}$ can vary along the axial length.

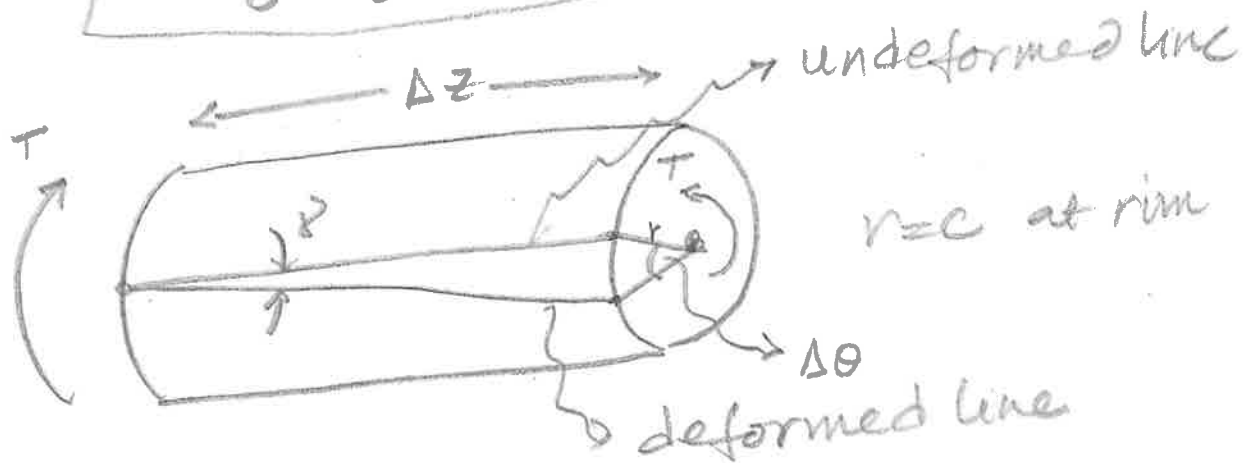
Due to twisting, unlike previous problem spatial dependence cannot be ignored

$$T = \int r \sigma_{z\theta} dA$$

$$= \int_0^c \int_0^{2\pi} \sigma_{z\theta} r^2 d\theta dr$$

⑦

→ ⑩



$$\tan \gamma_c = \frac{c \Delta\theta}{\Delta z}$$

$$\text{as } \Delta z \rightarrow 0 \quad \lim_{\Delta z \rightarrow 0} \frac{c \Delta\theta}{\Delta z} = c \frac{d\theta}{dz}$$

{ we made
small angle
assumption }

$$\tan \gamma_c \approx \gamma_c \Rightarrow \gamma_c = c \frac{d\theta}{dz} \text{ at } r=c$$

More generally,

$$\gamma(r) = r \frac{d\theta}{dz} \text{ at any } r$$

→ ⑪

From Hookean LEHI model
 $E_{z\theta}$ resists torque T

$$\textcircled{8} \quad \boxed{E_{z\theta} = \frac{1}{2G} \sigma_{z\theta}} \quad \textcircled{12}$$

We can also show $E_{z\theta} = \frac{1}{2} \gamma$ $\textcircled{or} \textcircled{12} \quad \boxed{\gamma = 2E_{z\theta}} \quad \textcircled{13}$

For $r=c$, $\gamma_c = 2E_{z\theta}(r=c) \rightarrow \textcircled{14}$

For any r , $\gamma_r = 2E_{z\theta}(r) \rightarrow \textcircled{15}$

Substituting $\textcircled{14}$, $\textcircled{15}$ into $\textcircled{12}$

$$\sigma_{z\theta}(r) = 2G E_{z\theta}(r) = G \gamma_r = G r \frac{d\theta}{dz} \rightarrow \textcircled{16}$$

$$\sigma_{z\theta}(c) = 2G E_{z\theta}(c) = G \gamma_c = G c \frac{d\theta}{dz} \rightarrow \textcircled{17}$$

$$\frac{\sigma_{z\theta}(r)}{\sigma_{z\theta}(c)} = \frac{r}{c} \Rightarrow \sigma_{z\theta}(r) = \frac{r}{c} \sigma_{z\theta}(c) \rightarrow \textcircled{18}$$

From $\textcircled{10}$ & $\textcircled{18}$ $T = \int r \sigma_{z\theta} dA$

$$\Rightarrow T = \int (r) \left(\frac{r}{c} \right) \sigma_{z\theta}(c) dA$$

$$\boxed{T = \frac{\sigma_{z\theta}(c)}{c} \int r^2 dA} \rightarrow \textcircled{19}$$

$\int r^2 dA$
 is numerical value

J is second polar moment of area

(9)

$$T = \frac{\tau_{z\theta}(c)}{c} J \Rightarrow \boxed{\tau_{z\theta}(c) = \frac{TC}{J}} \rightarrow (20)$$

$$\text{Hby } \therefore \tau_{z\theta}(r) = \frac{\tau_{z\theta}(c)}{c} r$$

$$= \frac{TC}{J} \frac{1}{c} r$$

$$\Rightarrow \boxed{\tau_{z\theta}(r) = \frac{Tr}{J}} \rightarrow (21)$$

$\Downarrow$ Universal result for stress
relative to (r, θ, z)
in terms of applied torque (T) .

Recall other universal results:

$$\sigma_{xx} = \frac{f}{A}$$

$$\sigma_{\theta\theta} = \frac{Pa}{h}$$

$$\sigma_{zz} = \frac{Pa}{2h} + \frac{f}{2\pi ah}$$

$$\sigma_{\theta\theta} = \frac{Pa}{2h} = \sigma_{\phi\phi}$$

Principal Stress & Strains in Torsion :

$$\sigma_{xx}' = \sigma_{xx} \cos^2 \alpha + 2\sigma_{xy} \sin \alpha \cos \alpha + \sigma_{yy} \sin^2 \alpha$$

Let $x \rightarrow z$; $y \rightarrow \theta$

$$\Rightarrow \sigma_{zz}' = \sigma_{zz} \cos^2 \alpha + 2\sigma_{z\theta} \sin \alpha \cos \alpha + \sigma_{\theta\theta} \sin^2 \alpha$$

In pure torsion:

where α is
angle between
 z & z'

$$\sigma_{zz} = \sigma_{\theta\theta} = 0$$

$$\therefore \sigma_{zz}' = 2\sigma_{z\theta} \cos \alpha \sin \alpha$$

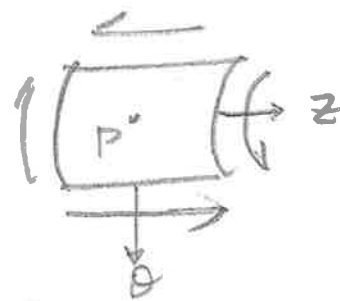
$$\Rightarrow \boxed{\sigma_{zz}' = 2 \frac{\tau_r}{J} \cos \alpha \sin \alpha} \text{ from (21)}$$

$\rightarrow$ (22)

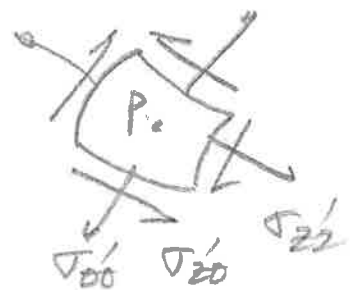
Wkly

$$\boxed{\sigma_{z\theta}' = \frac{\tau_r}{J} (\cos^2 \alpha - \sin^2 \alpha)}$$

$\rightarrow$ (22)



Only $\sigma_{z\theta} = \sigma_{\theta z} = \frac{\tau_r}{J}$



(11)

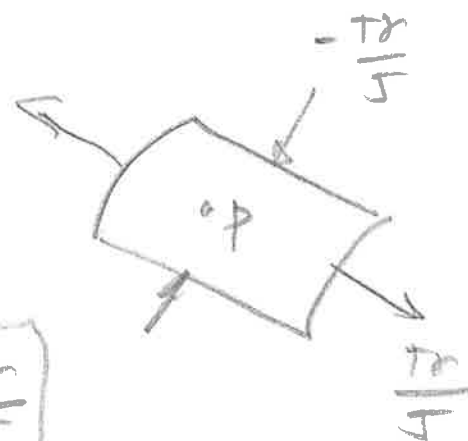
Recall

$$\sigma'_{22} \Big|_{\max, \min} = \frac{\sigma_{22} + \sigma_{00}}{2} \pm \sqrt{\left(\frac{\sigma_{22} - \sigma_{00}}{2}\right)^2 + \tau_{20}^2}$$

$$\sigma_{1,2} = \begin{cases} \sigma'_{22} \\ \sigma'_{00} \end{cases} \Big|_{\max, \min} = \frac{\sigma_{22} + \sigma_{00}}{2} \pm \sqrt{\left(\frac{\sigma_{22} - \sigma_{00}}{2}\right)^2 + \tau_{20}^2}$$

For pure torsion: $\sigma_{22} = \sigma_{00} = 0$

$$\sigma_{1,2} = \begin{cases} \sigma'_{22} \Big|_{\max, \min} = \pm \tau_{20} \\ \sigma'_{00} \Big|_{\max, \min} = \pm \tau_{20} \end{cases}$$



$$\Rightarrow \boxed{\sigma_1 = +\frac{\tau_r}{J}} \quad \& \quad \boxed{\sigma_2 = -\frac{\tau_r}{J}}$$

(25)

Plane on which max. normal stress acts

$$2\alpha_p = \tan^{-1} \left[\frac{\tau_{20}}{(\sigma_{22} - \sigma_{00})/2} \right]$$

$$\Rightarrow \alpha_p = \frac{1}{2} \tan^{-1}(\infty)$$

$$(\because \sigma_{22} = \sigma_{00} = 0)$$

$$\boxed{\alpha_p = \frac{\pi}{4} = 45^\circ}$$

Plane in which max. shear stress acts:

$$\alpha_s = \frac{1}{2} \tan^{-1} \left(\frac{\sigma_{\theta\theta} - \sigma_{zz}}{2\tau_{z\theta}} \right)$$

$$\Rightarrow \boxed{\alpha_s = \frac{1}{2} \tan^{-1}(0) = 0} \quad (\because \sigma_{\theta\theta} = \sigma_{zz} = 0)$$

↳ (24)

Insights:

- ① Largest max/min values occur at $r=c$
- ② Whether materials fails due to shear or normal stress, they will initiate at the outer surface ($r=c$)